

Mark schemes

Q1.

(a) $\mathbf{F} = (3\mathbf{i} + 4\mathbf{j}) + (6\mathbf{i} - 8\mathbf{j})$
Addition of the two forces

M1

2

$= 9\mathbf{i} - 4\mathbf{j}$
Correct resultant

A1

(b) $F = \sqrt{9^2 + 4^2} = 9.85 \text{ N}$
Finding magnitude

M1

Correct magnitude

A1

2

(c) $\tan \alpha = \frac{4}{9}$
Using tan to find the angle

M1

Correct equation

A1

$\alpha = 24.0^\circ$
Correct angle

A1

3

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Q2.

(a) $\mathbf{F} = (2 + 6 - 2)\mathbf{i} + (3 - 8 + 10)\mathbf{j}$
Summing the forces

M1

$= 6\mathbf{i} + 5\mathbf{j}$
Correct resultant

A1

2

(b) $\mathbf{a} = \frac{1}{8} (6\mathbf{i} + 5\mathbf{j}) = 0.75\mathbf{i} + 0.625\mathbf{j}$
Dividing by 8 to get correct value

B1

1

(c) (i) $\mathbf{v} = (3\mathbf{i} - 2\mathbf{j}) + 40(0.75\mathbf{i} + 0.625\mathbf{j})$
use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$

M1

$= 33\mathbf{i} + 23\mathbf{j}$
Correct $\mathbf{v}$

A1

2

(ii) $\mathbf{r} = (3\mathbf{i} - 2\mathbf{j}) \times 40 + \frac{1}{2} (0.75\mathbf{i} + 0.625\mathbf{j}) \times 40^2$
Vector constant acceleration equation to find $\mathbf{r}$
Correct equation

M1 A1

$= 720\mathbf{i} + 420\mathbf{j}$
Correct position vector

A1

$r = \sqrt{720^2 + 420^2} = 834 \text{ m}$

Finding magnitude
Correct magnitude

M1 A1

5

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Q3.

(a) $44\mathbf{i} + 28\mathbf{j} = 10(4\mathbf{i} + 2\mathbf{j}) + 50\mathbf{a}$
Equation to find $\mathbf{a}$ using position vector
Correct equation

M1/A1

$\mathbf{a} = 0.08\mathbf{i} + 0.16\mathbf{j}$
Solving for $\mathbf{a}$
Correct $\mathbf{a}$

M1/A1

4

(b) $\mathbf{r} = (4\mathbf{i} + 2\mathbf{j})t + \frac{1}{2}(0.08\mathbf{i} + 0.16\mathbf{j})t^2$
Equation for r
One component correct

M1/A1

$= (4t + 0.04t^2)\mathbf{i} + (2t + 0.08t^2)\mathbf{j}$
Second component correct

A1

3

(c) (i) $96 = 4T + 0.04T^2 \quad 72 = 2T + 0.08T^2$
Forms and solves equation for one component

M1

$T = 20 \text{ or } -120 \quad T = 20 \text{ or } -45$
Correct solutions

A1

Hence $T = 20$
Forms and solves equation for other component
Correct solutions plus correct conclusion

M1/A1

4

(d) (i) $\mathbf{v} = (4\mathbf{i} + 2\mathbf{j}) + (0.08\mathbf{i} + 0.16\mathbf{j}) \times 20$
Finding v at t = 20

M1

$= 5.6\mathbf{i} + 5.2\mathbf{j}$
Correct v

A1

$v = \sqrt{5.6^2 + 5.2^2} = 7.64 \text{ ms}^{-1}$
Finding correct speed

A1

3

[14]